NFL-Auditor

Evidence-Bound Scientific Reviewing with Deterministic Auditing and a Grounded Multi-Agent Committee

Abstract

Automated paper review must combine scientific judgment with resistance to fabricated evidence, prompt injection, and operational failure. NFL-Auditor is a hybrid Track 2 review system for evaluating Ralphthon Track 1 papers in the ICML review format. Its deterministic layer converts each PDF into a canonical, content-addressed document; quarantines concealed payloads before analysis; extracts claims, tables, citations, and numeric results; and runs reproducible consistency, arithmetic, evidence, positioning, and integrity checks. Scientific quality is assessed by three concurrent specialists: a theorist for problem-method fit, an experimentalist for claim-experiment alignment and validity, and a scope/ablation reviewer for generalization and design choices. A fourth area-chair stage reconciles their structured assessments into grounded strengths, weaknesses, author questions, and all six platform scores. Strict schema and evidence validation reject unsupported output, while deterministic findings and integrity caps remain authoritative. A guidance-driven API adapter validates the fixed ten-paper assignment set, preserves local artifacts, and separates review preparation from time-gated submission.

1. System Objective

A mechanical checker can prove an arithmetic error but cannot decide whether a method addresses its stated problem. A free-form model can discuss research quality but may invent evidence or follow instructions embedded in a paper. NFL-Auditor combines both capabilities while keeping their responsibilities separate: deterministic code establishes auditable facts, and a validated committee supplies scientific judgment.

2. End-to-End Architecture

Secure PDF ingestion	Deterministic S1-S6 audit	3 scientific specialists	Area-chair meta-review	Validated review and API payload

The architecture is governed by five implementation principles:

1. **Sanitize before interpretation:** hidden content is removed before parsing, retrieval, scoring, or model use.
2. **Evidence before verdict:** central comments and score rationales cite stable evidence identifiers.
3. **Separate identities:** deterministic audit and scientific judgment provenance are frozen independently.
4. **Specialize scientific review:** logical fit, experiments, and scope receive independent assessments.
5. **Isolate failures:** one paper or service failure cannot cancel the remaining review batch.

3. Canonical Document and Injection Resistance

The original PDF and derived Markdown are represented as distinct immutable sources. The canonical record captures media type, byte length, page count, converter version, and content digest. The local path is provenance only and is excluded from the content-addressed review identity.

3.1 Concealed PDF text quarantine

Ordinary PDF conversion can flatten white-on-white, transparent, or non-rendering text into normal extracted text. NFL-Auditor inspects PDF spans before conversion, estimates the page background, and quarantines low-contrast, transparent, sub-pixel, and non-rendering spans. Concealed text never reaches claim extraction, findings, scores, or scientific review. Location-only integrity records remain available for the audit.

3.2 Sanitize-first text pipeline

The Markdown sanitizer removes hidden HTML, comments, Unicode format controls, and reviewer-directed instructions while preserving line structure. The parser then inventories sections, paragraphs, tables, references, numeric tokens, and line/column locations. Scientific evidence is split into bounded stable spans such as `paper:L120–L131`.

4. Deterministic S1-S6 Audit

1. **S1 Parse:** construct section, table, citation, and numeric-token inventories.
2. **S2 Claims:** extract falsifiable result, arithmetic, hypothesis, and declarative claims.
3. **S3 Checks:** execute evidence, consistency, formatting, positioning, and integrity checks.
4. **S4 Verdicts:** label claims supported, contradicted, or unverifiable with evidence pointers.
5. **S5 Compose:** draft comments and retain only comments licensed by their evidence.
6. **S6 Freeze:** bind source identities, evidence snapshots, agent version, and verdict labels.

The check battery covers result-ledger traceability, table-to-prose consistency, arithmetic recomputation, baseline fairness, omitted negative outcomes, citation existence, template compliance, related-work positioning, self-review consistency, and injection scanning. Findings are matched to compatible claim types and overlapping spans rather than indiscriminately affecting every claim on a line.

5. Evidence-Grounded Composition

Composition uses a draft-and-ground pattern. Unsupported praise is removed; criticism without defect evidence becomes an author question. An empty References heading is not treated as positioning, and an unavailable external lookup is recorded as unavailable rather than as a paper defect. The audit therefore distinguishes “not verified” from “proven false.”

> The deterministic review identity and ordered verdict digest are stable records of what the code established from the frozen input.

6. Scientific Review Committee

The scientific layer is a real four-stage committee. Three specialist calls run concurrently over role-targeted packets assembled only from sanitized paper spans and deterministic findings.

Theorist

Assesses problem formulation, assumptions, logical validity, and whether the proposed method addresses the stated problem.

Experimentalist

Assesses claim-experiment alignment, controls, baselines, metrics, confounds, statistical support, and experimental validity.

Scope and ablation reviewer

Assesses generalization, experimental scope, design-choice justification, ablations, robustness, and stated limitations.

7. Area-Chair Synthesis

When at least two specialists produce valid assessments, an area-chair stage reconciles their outputs with the deterministic audit and retrieved prior-work records. The final structured judgment covers five scientific axes: problem-method fit, claim-experiment alignment, experimental-design validity, scope/generalization, and design-choice/ablation justification.

The synthesis produces a paper-specific summary, grounded strengths and weaknesses, three to five numbered author questions, and an explanation of what each answer would change. Every assessment and score rationale must cite an allowed paper span, claim, finding, or literature record. Missing axes, unknown evidence IDs, duplicate IDs, empty text, extra fields, incorrect numeric types, and out-of-range scores are rejected.

8. Direct Platform Scoring

Dimension	Range	Primary basis
Soundness	1-4	Logical and empirical validity
Presentation	1-4	Clarity and completeness
Significance	1-4	Importance of supported contribution
Originality	1-4	Positioning and differentiated contribution
Overall	1-6	Integrated recommendation
Confidence	1-5	Coverage and evidential certainty

Valid scientific scores may move above or below deterministic anchors. Mechanical contradictions remain visible, and proven contradictions, dishonest self-certification, or concealed-content findings cap Soundness and Overall at 2 after scientific scoring.

9. Failure-Isolated Review

One failed specialist does not cancel synthesis. If specialist quorum is unavailable or the area-chair result is invalid, the paper receives the complete deterministic review. Model, retrieval, or conversion failures are isolated to their assignment, allowing the remaining papers to proceed.

10. Guidance-Driven Ten-Paper Workflow

The live adapter treats the organizer's canonical `skill.md` as the operational contract. It parses typed stage, reason, actor, prerequisite, next-action, and KST timing values and branches on stable reason codes rather than prose.

1. Exchange a human-issued setup token once; retain the bearer only in memory.
2. Read authenticated status without implicitly allocating work.
3. Acquire assignments only when returned guidance permits it.
4. Validate exactly ten unique ordinals, 1-10, before optional filtering.
5. Download only returned scoped HTTPS PDF URLs on the canonical host.
6. Store originals under `submit_work/assigned_papers/ordinal-XX/paper.pdf`.
7. Prepare paper reviews concurrently before the write window.
8. Refresh status and canonical guidance before each serialized review POST.
9. Submit only in the half-open interval [16:35, 17:00) KST and stop on terminal guidance.

11. Output and Reproducibility

Each local review contains Summary, Strengths, Weaknesses, Questions for the Authors, Scores, Ethics and Limitations, Evidence Trace, and a closing recommendation. The local trace records original and derived identities, citation snapshots, deterministic verdicts, committee call provenance, and the independent scientific judgment identity.

The live payload contains exactly the platform fields: ordinal, Soundness, Presentation, Significance, Originality, Overall, Confidence, and comments. Before transmission, NFL-Auditor extracts only substantive review sections; local paths, content hashes, prompt hashes, storage details, and private assignment metadata remain local.

12. Operational Safety

- Credentials are not accepted through command-line arguments and are never written to disk.
- Redirects are rejected before authenticated downloads.
- Prepared reviews are reused only when source, agent, mode, schema, and artifact identities match.
- Already-submitted ordinals are skipped, enabling safe resume.
- Transient service errors use bounded retries; permanent validation errors are not repeated unchanged.

13. Conclusion

NFL-Auditor combines a reproducible evidence and integrity substrate with specialized scientific reasoning. The deterministic layer protects the review from concealed content, unsupported claims, arithmetic errors, and service failure. The committee supplies the problem-method, experimental, scope, and ablation judgments that a mechanical audit cannot provide. Strict validation, independent identities, hard integrity caps, local artifact preservation, and per-paper fallback support a time-bounded ten-paper Ralphthon workflow.

Implementation Map

Secure ingestion: `document.py`, `to_markdown.py`, `injection_scan.py`. Audit and composition: `pipeline.py`, `claims.py`, `composer.py`. Scientific committee: `scientific_review.py`, `model_critique.py`, `review_schema.py`. Deployment: `agent_api.py`, `api_scores.py`, and `submit.py`.